# MILITARY SPECIFICATION

## SHELTER, ELECTRICAL EQUIPMENT S-301/TCC-61

### 1. SCOPE

1.1 Scope.- This specification covers the air or transportable shelter facility part of the Terminal, Telephone, AN/TCC-61() which is capable of providing multiplex terminal facilities of eight (8) groups of twelve (12) channels or four (4) groups of twenty-four (24) channels for cable transmission or, with other equipments for radio transmission.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-C-3885 | Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment) |
| MIL-W-6858 | Welding, Aluminum, Magnesium, Non-Hardening Steels or Alloys and Titanium, Spot, Seam and Stitch |
| MIL-T-7928 | Terminals Lug and Splice, Crimp Style, Copper |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finishes for Ground Signal Equipment |
| MIL-P-55505 | Shelter, Electrical Equipment (Containing Equipment) Packaging and Packing of |

FSC 5805

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-202 | Test Methods for Electronic and Electrical Component Parts |
| MIL-STD-252 | Wired Equipment, Classification of Visual and Mechanical Defects |
| MIL-STD-454 | Standard General Requirements |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |
| MIL-STD-1235 | Single, Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

DRAWINGS

ELECTRONICS COMMAND

| | |
|---|---|
| DL-SC-B-427758 | Shelter, Electrical Equipment S-301()/TCC-61 |
| SC-D-427801 | Power Schematic Diagram S-301/TCC-61 |
| SC-A-46691B | R.F.Cable Assembly Tests |
| SC-C-532288 | Schematic Diagram, Voltage Cutout Protector |
| SC-D-681007 | Cable Assembly Special Purpose Electrical |
| SC-D-681008 | Signal Entrance Assembly, Wiring Diagram |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number of symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Description.- Shelter, Electrical Equipment S-301()/TCC-61 hereinafter called the shelter facility, includes a modified S-280()/G Shelter and all racks, brackets, shelves, tie-downs, electrical components and wiring to enable installation and operation of the assemblage components of the AN/TCC-61() See 6.1.

3.2 Preproduction sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one preproduction sample of the shelter facility for approval. (See 4.3)



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3.3 Construction.- The shelter facility shall be constructed in accordance with drawings as listed on DL-SC-B-427758.

3.4 Electrical installation.-

3.4.1 Power system wiring and components.- Installation of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly and wiring components constituting a three wire electrical power distribution system) shall be installed and wired in accordance with the drawings. The completed in-place power distribution system shall meet the following requirements:

3.4.1.1 Continuity of conductors.- Electrical continuity shall exist between each of the pins numbered 1,2, and 4 (of power connector J1 in power entrance box) and its respective circuit terminations (at receptacles, switches, lighting fixtures and lamp holders) and also between each of the pins numbered 1,2, and 4 (of power connector J1) and its respective termination (at pins numbered 1,2, and 4 of power connector J2) when tested in accordance with 4.6.1 and 4.6.1.1.

NOTE: Tests to satisfy the requirements of 3.4.1.1 (See 4.6.1 and 4.6.1.1) shall be accomplished following completion of power system wiring and prior to mounting of AC power receptacles in the wireway.

3.4.1.2 Dielectric strength of insulation.- The insulation of all wires, cables, and components shall withstand a potential of 600 volts DC continuously applied for not less than 30 seconds when tested in accordance with 4.6.1 and 4.6.1.2.

3.4.1.3 Insulation resistance.- The resistance of insulation between any two of the three circuit conductors and also between each of the three circuit conductors and shelter ground shall not be less than 100 megohms when tested in accordance with 4.6.1 and 4.6.1.3.

3.4.2 Overvoltage control.- The overvoltage control shall be fabricated and wired in accordance with the drawings and shall meet the following requirements:

3.4.2.1 Dielectric strength of insulation.- As required in 3.4.1.2 when tested in accordance with 4.6.2.1.

3.4.2.2 Insulation resistance.- As required in 3.4.1.3 when tested in accordance with 4.6.2.2.

3.4.2.3 Voltage drop.- A voltage drop of less than 250 mv shall exist between each pin of connector P3 and its respective pin of connector P4 when the circuit current is 50 amperes. (See 4.6.2.3.)



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3.4.3 Power distribution and operational functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as denoted by power system schematic drawing SC-D-427801 when tested in accordance with 4.6.3.

3.4.4 Signal (audio) system wiring and components.- Installation of signal (audio) wires, cables and wiring components (including binding posts, connectors, plugs, jacks, receptacles, switches, terminal strips, terminal blocks, and wiring devices) shall be installed and connected in accordance with the drawings. The completed in-place signal (audio) system installation shall meet the following requirements:

3.4.4.1 Continuity and resistance.- There shall be circuit continuity between the following wire terminations (resistance of these continuous circuits shall not exceed 5 ohms) when tested in accordance with 4.6.4.1:

a. Pins 1A thru 24B of each audio connector J1 thru J8 and the corresponding pins of audio connectors systems 1 thru 8 when the respective CABLE/BINDING POST channel switch is placed in the CABLE position. (See drawings SC-D-681008).

b. Binding posts 1A thru 24B of each system (1 thru 8) and the corresponding pins of audio connectors of systems 1 thru 8, when the respective CABLE/BINDING POST channel switch in placed in the BINDING POST position.

c. Each phone jack terminal and each of the following: corresponding binding posts in the signal entrance box, corresponding pins of audio connectors (J1 thru J8), and corresponding binding posts in the power entrance box.

d. Each intercom jack terminal and each of the following: corresponding binding posts in the signal entrance box, corresponding pins of audio connectors (J1 thru J8) and the corresponding binding posts in the power entrance box.

3.4.4.2 Dielectric strength of insulation.- The insulation of all wires, cables, and components shall withstand a potential of 600 volts DC continuously applied for not less than 5 seconds when tested in accordance with 4.6.4.2.

3.4.4.3 Insulation resistance.- The resistance between each of the following shall not be less than 100 megohms when tested in accordance with 4.6.4.3:

a. Each wire conductor and all remaining wire conductors.

b. Each component conductor and all remaining component conductors.



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c. Each wire conductor and shelter ground.

d. Each component conductor and shelter ground.

e. Each wire conductor and connector shields.

3.4.5 Video system wiring and components.- Installation of video cables and connecting components (including plugs, connectors, adapters, fittings, receptacles, and lightning protectors) shall meet the following requirements:

3.4.5.1 Continuity and Circuit resistance.- There shall be circuit continuity between the following video cable terminations (resistance of the continuous circuit shall not exceed 3 ohms when tested in accordance with 4.6.5.1):

a. Both end terminations of the inner conductor.

b. Both end terminations of outer conductor (shield).

c. Each end of the outer conductor and facility ground terminal.

3.4.5.2 Dielectric strength of insulation.- The insulation of all video cables, components, and lightning protectors shall withstand the following potentials when tested in accordance with 4.6.5.2:

a. 1700 volts DC (applied continuously for not less than 30 seconds) to all video cables with lightning protector elements removed.

b. 1200 volts DC (applied continuously for not less than 30 seconds) to all video cables with lightning protector elements in place.

3.4.5.3 Insulation resistance.- The resistance between each inner conductor and outer conductor (shield) and between each inner conductor and shelter ground shall not be less than 100 megohms when tested in accordance with 4.6.5.3.

3.4.6 Cable Assemblies.- (See 4.4)

3.4.6.1 Cable Assemblies, Power and audio frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the test requirements of MIL-C-3885.



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3.4.6.2 Cable Assemblies Radio Frequency.- Radio frequency cable assemblies required for operation of major items of equipment in 6.1 shall be fabricated in accordance with the drawings and meet the test requirements of drawing SC-A-46691B.

3.4.7 Suppression of radio frequency interference.- The motors in the ventilating equipment used in the facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items specified on the drawings. (See 4.4)

3.5 System operation.- With all operational equipment of the AN/TCC-61() assemblages installed, and system 1 equipment connected to system 5 equipment, system 2 equipment connected to system 6 equipment, system 3 equipment connected to system 7 equipment and system 4 equipment connected to system 8 equipment connected back to back, communications shall be established between systems. (See 4.7)

3.6 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicles without sustaining any permanent damage. (See 4.8)

3.7 Railroad transportation.- The shelter facility shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail. When so loaded, the shelter facility shall be subjected to and withstand the test of paragraph 4.9 without sustaining any buckling, delamination, or other permanent deformation.

3.8 Drops, flat and rotational.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete. After all drop tests of 4.10.1 and 4.10.2 there shall be no buckling or delamination or other permanent deformation.

3.9 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.11. After the test there shall have been no leakage into the facility.

3.10 Fording.- The shelter facility shall be capable of being immersed in water to a depth of 21 inches, measured from the bottom of the skids, for 1 hour without any leakage of water into the facility. (See 4.12)

3.11 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and when installed each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.13. There shall be no damage to the insert threads, the insert shall not have rotated from its original position, and there shall be no permanent distortion of the shelter mounting surface or insert head or threads. Insert threads and outer face of insert head shall be free of paint.



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3.12 Cleaning.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary such cleaning shall also be performed before and after final assembly of the units.

3.13 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the drawings.

3.14 Marking.-

3.14.1 General.- Marking shall conform to Specification MIL-M-13231 and the equipment drawings.

3.14.2 Aircraft loading data plate.- Each shelter facility shall be provided with data plates as indicated on the drawings.

3.14.3 Nameplate.- Each shelter facility shall be serial numbered and provided with a nameplate in accordance with the drawings.

3.14.4 Interior and exterior marking.- All interior and exterior marking of the shelter facility shall be as depicted on the drawings.

3.15 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.16 Seal vapor.- All joints and raw edges of the framing for the power, signal and video entrance boxes, and exhaust blowers shall be sealed to prevent entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints shall be sealed in a manner to prevent collection and retension of moisture. All exterior rivets and screws shall be dipped in sealer prior to installation. The sealer shall be one part Silicone Construction Sealant SE 1204 as made by General Electric Co. or equal.

3.17 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.17.1 Screws, threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.



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3.18 Soldering.-

3.18.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.18.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.18.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, over-heated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.19 Tropicalization of material.- Canvas, webbing, wood, etc., shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

3.19.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.19.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.19.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.19.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use (See 6.2e)



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3.20 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary except where grinding is specified on the drawings. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858, Class B. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

3.21 Wiring and cabling.- Wiring and cabling shall be in accordance with the drawings and applicable portions of 9 & 19 of MIL-STD-454.

3.21.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.21.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.21.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.21.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests Specification MIL-T-7928. (See 4.4 and 4.14)

3.21.5 Identification of wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.



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3.22 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.15) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.23 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.3 Construction
3.4 Electrical installation
3.11 Mounting inserts
3.12 Cleaning
3.13 Finish
3.14 Marking
3.15 Riveting
3.16 Seal, vapor
3.17 Securing of parts
3.18 Soldering
3.19 Tropicalization
3.20 Welding
3.21 Wiring and cabling
3.22 Interchangeability

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents.(See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.17)



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4.3 Preproduction inspection.- This inspection will be performed by the contractor on the first article unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (see Tables II,III, and IV, respectively). The preproduction inspection will normally be performed in this order:

a. Mounting inserts
b. Power wiring
c. Signal wiring
d. System operation
e. Vehicular transportation
f. Rail transportation
g. Flat drop
h. Rotational drop
i. Watertightness
j. Fording

The last tests performed shall be a repeat of the power, signal, and system tests. The remaining preproduction tests shall be performed in any order satisfactory to the contracting officer.

Table I - Preproduction Inspection

| Inspection (For additional preproduction inspection see 4.3) | Rqr Para | Insp Para |
|---|---|---|
| System operation | 3.5 | 4.7 |
| Vehicular transportation | 3.6 | 4.8 |
| Rail transportation | 3.7 | 4.9 |
| Mounting inserts | 3.11 | 4.13.1 |



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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

| Item | Where Required |
|---|---|
| Cable assemblies | 3.4.6 |
| Suppression, RFI | 3.4.7 |
| Finish | 3.13 |
| Marking | 3.14 |
| Welding | 3.20 |
| Terminals | 3.21.4 |

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements; (See 6.6). Test equipment for government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105 using the general inspection levels.



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## Table II - Group A Inspection

| Inspection | Rqr Para | Insp Para | AQL Major | AQL Minor |
|---|---|---|---|---|
| Visual and Mechanical | 3.23 | 4.16 | 1.5 dphu* | 6.5 dphu* |
| Electrical | | | | ** |
| Power Wiring | | | | |
| Continuity | 3.4.1.1 | 4.6.1.1 | | |
| Dielectric strength | 3.4.1.2 | 4.6.1.2 | | |
| Insulation resistance | 3.4.1.3 | 4.6.1.3 | | |
| Overvoltage control | | | | |
| Dielectric strength | 3.4.2.1 | 4.6.2.1 | | |
| Insulation resistance | 3.4.2.2 | 4.6.2.2 | | |
| Voltage drop | 3.4.2.3 | 4.6.2.3 | | |
| Power distribution | 3.4.3 | 4.6.3 thru 4.6.3.3 | 1% for the entire group of power and signal wiring | |
| Signal wiring | | | | |
| Continuity & circuit resistance | 3.4.4.1 | 4.6.4.1 | | |
| Dielectric strength | 3.4.4.2 | 4.6.4.2 | | |
| Insulation resistance | 3.4.4.3 | 4.6.4.3 | | |
| Video system | | | | |
| Continuity & circuit resistance | 3.4.5.1 | 4.6.5.1 | | |
| Dielectric strength | 3.4.5.2 | 4.6.5.2 | | |
| Insulation resistance | 3.4.5.3 | 4.6.5.3 | | |

*Defects per hundred units
** All defects are major.



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4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table III shall be as follows:

| AQL | Inspection level |
|---|---|
| 4.0% | S-4 |
| 6.5% | S-4 |

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed on the same equipment in the order in which they appear.

Table III - Group B Inspection

| Inspection | Rqr Para | Insp Para | AQL (see 4.5.2.1) |
|---|---|---|---|
| Mounting insert | 3.11 | 4.13.2 | 4.0% |
| Watertightness | 3.9 | 4.11 | 6.5% |
| Fording | 3.10 | 4.12 | 6.5% |
| Interchangeability | 3.22 | 4.15 | 4.0% |

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

Table IV - Group C Inspection

| Inspection | Rqr Para | Insp Para |
|---|---|---|
| Drops, flat and rotational | 3.8 | 4.10 |
| Solderless-terminal-lug connection | 3.21.4 | 4.14 |



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4.5.3.1 Sampling for inspection of drop tests.- One sample shall be selected at random from the first 50 units or fraction thereof, and one sample from each additional 150 units or fraction thereof, produced and subjected to the drop tests.

4.5.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units or product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative.
After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the government) and Groups A and B inspection may be reinstituted; final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.4)

4.5.4 Reinspection of conforming Group B and Group C sample units.- Inless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group C inspection after repair of all visible damage.

4.5.5 Alternate inspection procedures.- Contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plan stated in 4.5.1 and 4.5.2. If this alternate is selected the procedure of implementation must be in accordance with MIL-STD-1235, "Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes". Before starting production Contractor shall notify the contracting office in writing which method he intends to use. The method may not be changed during production without written approval of the Government.

4.6 Electrical installation tests.- The tests of 4.6.1 through 4.6.5.3b shall be performed on each shelter facility prior to installation of other major components comprising the AN/TCC-61 or dummy loads. Before applying power to the shelter facility, an electrical ground at the Power Entrance Box shall be made. Power to the shelter facility shall be 115 VAC, 60 cycles, single phase.



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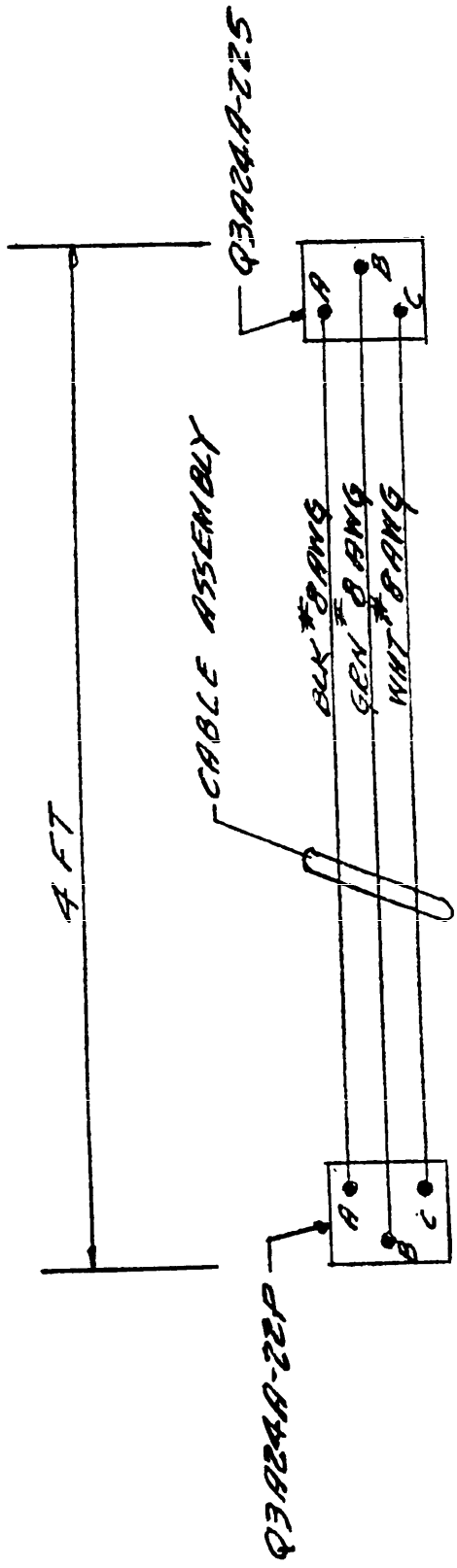



FIG. 1



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4.6.1 Power system and components installation tests.- Prior to conducting power wiring and components tests of 4.6.1.1 and 4.6.1.2, and 4.6.1.3, the following shall be accomplished: (See Dwg SC-D-427801)

a. Remove or disconnect the following electrical loads:

(1) Fluorescent lamps and lamp starters, DS13 thru DS20.

(2) Incandescent cold start lamps, DS21 thru DS24.

(3) Power distribution boxes neon indicator lamps, DS1 thru DS7, DS8 thru DS11.

(4) Power indicator neon lamp DS12.

(5) Blower-1 and blower-2.

b. Remove power distribution box cover panel and disconnect both conductors from the voltmeter terminals. Position or isolate disconnected leads so that they do not short to each other or to shelter ground.

c. Fabricate test cable assembly shown in FIGURE 1. Connect test cable between power cable marked "Input to CN-514," and power cable marked "Output Overvoltage Control."

d. Plug in cable from rack equipment raceway to associated receptacle P5 to J11, P6 to J12, P7 to J13 and P8 to J14.

4.6.1.1 Continuity of conductors.- Continuity measurements which satisfy the requirements 3.4.1.1 shall be made on each power-line circuit as indicated by the following paragraphs using as aural or visual indicating continuity test instrument and suitable test jig. The instrument shall produce a test current between 0.2 and 2.0 amperes AC when circuit continuity exists.

a. Remove and extend all connected power receptacles (J4 thru J34) from wireway so that the outlet mounting brackets do not make electrical contact to the wireway. (See FIGURE 2)

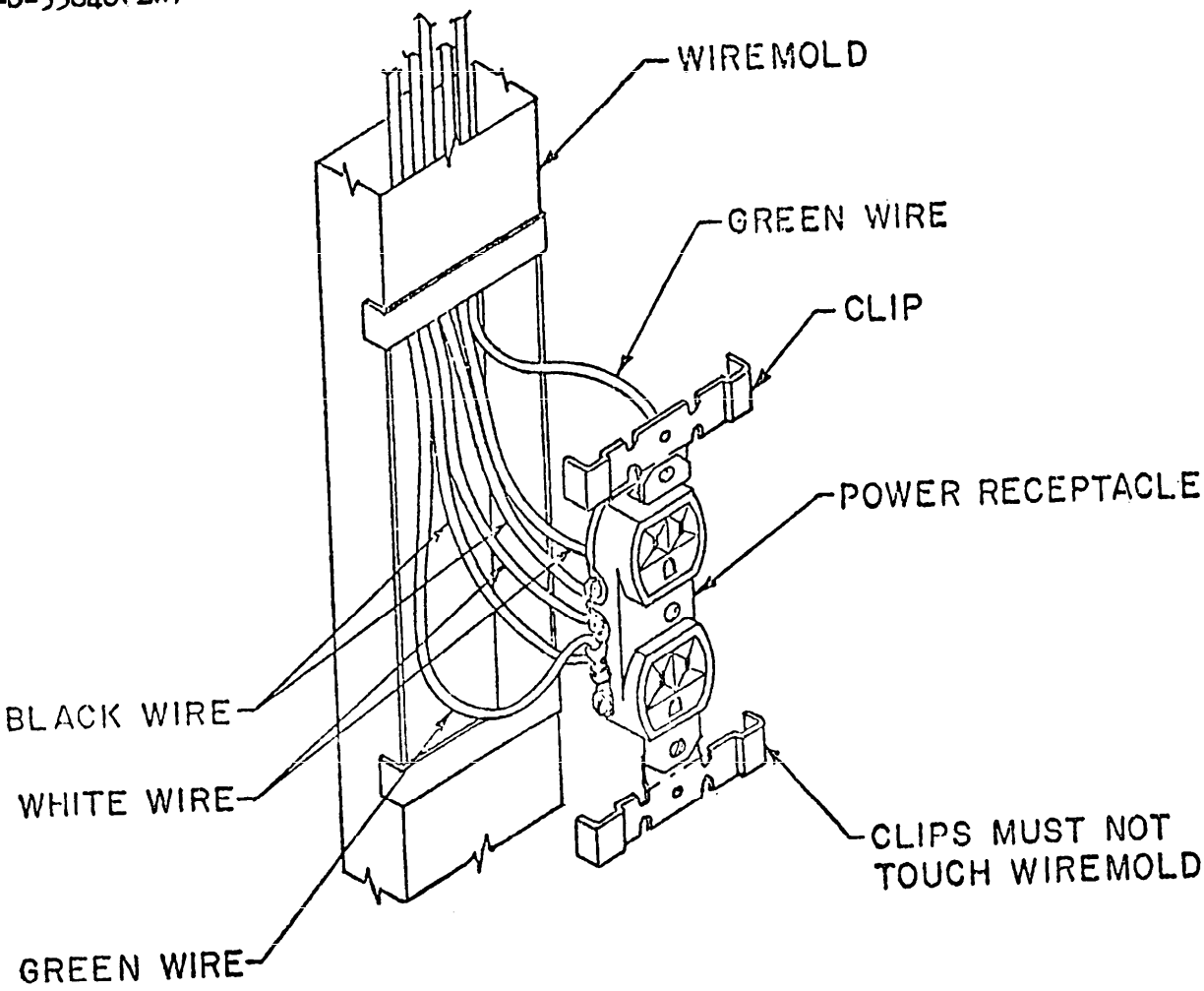
b. Place the following circuit controls in the ON position:

(1) Fluorescent lamp switches (S3) (S2)

(2) Incandescent cold start switches (S7)(S8)



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POWER RECEPTACLE REMOVED & EXTENDED FROM WIREMOLD

FIGURE NO. 2



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(3) Bypass blackout switch (S4)

(4) Main circuit breaker (CB8)

(5) Branch circuit breakers (CB1 thru CB7 and CB13)

(6) System circuit breakers (CB9 thru CB12)

(7) Blower switches (S5 and S6)

c. Measure continuity from Pin 1 (of power connector J1 in power entrance box) to the following: brass contact on each receptacle of J3 thru J34; black wire termination at each fluorescent lamp (DS13 thru DS20); and black wire termination at each incandescent lamp (DS21 thru DS24).

d. Place all controls listed under procedure 4.6.1.1b in OFF position, except main circuit breaker CB8.

e. Measure continuity from Pin 2 (of power connector J1) to the following: silver contacts on each receptacle of J4 thru J34; white wire termination at each fluorescent lamp (DS13 thru DS20) and white wire termination at each incandescent lamp (DS21 thru DS24).

f. Place main circuit breaker CB8 in OFF position.

g. Measure continuity from Pin 4 (ground) (of power connector J1) to each ground terminal on receptacles J3 thru J34.

h. Measure continuity from Pin 4 (ground) (of power in connector J1) to shelter ground terminal located in power entrance box panel.

i. Measure continuity from Pin 1 of IN connector J1 to Pin 1 of OUT connector; J2 from Pin 2 of IN connector J1 to Pin 2 of OUT connector J2; and from Pin 4 of IN connector J1 to Pin 4 of OUT connector J2.

j. Replace and secure power receptacles removed by procedure 4.6.1.1a.

k. Measurements made by 4.6.1.1c,e,g,h and i shall indicate existence of electrical circuit continuity. (See 3.4.1.1)

4.6.1.2 Dielectric strength of insulation.- Dielectric strength of insulation tests which satisfy the requirements of 3.4.1.2 shall be made on each power-line circuit indicated by the following paragraphs (Method 301 of MIL-STD-202, shall be employed using a nondestructive DC dielectric tester):



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a. Place all circuit controls listed in procedure 4.6.1.1b in the ON position.

b. Connect tester between Pin 1 and Pin 2 of power IN connector J1 in power entrance box. Adjust tester for a value of 600 volts DC and maintain this potential for a period of 30 seconds. The requirements of 3.4.1.2 shall be met.

c. Repeat procedure 4.6.1.2b with tester connected between Pin 1 and Pin 4 of power IN connector J2 except place S2 and S3 in OFF position.

d. Repeat procedure 4.6.1.2b with tester connected between Pin 2 and Pin 4 of power IN connector J2.

NOTE: To Eliminate danger of shock.- Discharge conductors following each test by connecting a 10 to 60 ohm wirewound resistor (or equivalent test lamp) to the pins tested.

4.6.1.3 Insulation resistance.- Insulation resistance tests which satisfy the requirements of 3.4.1.3, shall be made on each power-line circuit indicated by the following paragraph (Method 302 of MIL-STD-202 shall be employed using a megohmmeter or equivalent tester):

a. Connect tester between Pin 1 and Pin 2 of power connector J1 in power entrance box. Adjust tester for a value of not less than 100 nor more than 600 volts DC. Maintain this potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.4.1.3 shall be met.

b. Repeat procedure 4.6.1.3a with tester connected between Pin 1 and Pin 4 of power connector J1.

c. Repeat procedure 4.6.1.3a with tester connected between Pin 2 and Pin 4 of power connector J1.

NOTE: To Eliminate danger of shock: Discharge conductors following each test by connecting a 10 to 60 ohm wirewound resistor (or equivalent test lamp) to the pins tested.

4.6.2 Overvoltage control tests.- Verification of the electrical requirements of 3.4.2 shall be made by measurements as indicated in 4.6.2.1, 4.6.2.2, and 4.6.2.3, referenced to drawing SC-C-532288.



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4.6.2.1 Dielectric strength of insulation.-

a. Employing method 301 of MIL-STD-202, connect a nondestructive DC dielectric tester between PIN A and PIN C of connector P3. Adjust tester for a value of 600 volts DC and maintain this potential for a period of 30 seconds. The requirements of 3.4.2.1 shall be met.

b. Repeat procedure 4.6.2.1a with tester connected between Pin A and Pin B of connector P3.

c. Repeat procedure 4.6.2.1a with tester connected between Pin B and Pin C of connector P3.

d. Repeat procedure 4.6.2.1a,b, and c for connector P4 Pins A,B, and C.

4.6.2.2 Insulation resistance.-

a. Employing method 302 of MIL-STD-202, connect a DC megohmmeter or equivalent tester between Pin A and Pin C of connector P3. Adjust tester for a value of not less than 100 nor more than 600 volts DC and maintain this potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.4.2.2 shall be met.

b. Repeat procedure 4.6.2.2a with tester connected between Pin A and Pin B of connector P3.

c. Repeat procedure 4.6.2.2a with tester connected between Pin B and Pin C of connector P3.

d. Repeat procedure 4.6.2.2a,b,and c for connector P4 Pins A,B and C.

4.6.2.3 Voltage drops.-

a. Connect 115 volts AC (nominal), 60 cycles, across Pins A and C of connector P3.

b. Connect a resistive load across pins A and C of connector P4 to produce a current of approximately 50 $\pm$ 5 amps.

c. Place switch on overvoltage control unit in ON position. Neon lamp on overvoltage control unit shall light.

d. Measure voltage drop between Pin A of connector P3 and Pin A of connector P4. The requirements of 3.4.2.3 shall be met.



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e. Repeat procedure 4.6.2.3d for Pin C of connector P3, and Pin C of connector P4.

4.6.3 Power distribution and operational functions tests.- Power distribution and operational functions tests which satisfy the requirements of 3.4.3 shall be made as indicated by the following paragraphs:

4.6.3.1 Test preparation.- (See drawing SC-D-427801)

a. Replace or reconnect all electrical loads and line voltmeter removed or disconnected by procedure 4.6.1a and b.

b. Replace power distribution box cover.

c. Remove test cable assembly installed by 4.6.1c. Connect power cable connector marked "Input to CN-514" to INPUT connector and power cable connector marked"OUTPUT Overvoltage Control: to OUTPUT connector of overvoltage control unit.

d. Place circuit breakers (CB1 thru CB12) and switches (S2,S3, S7,S8 in the OFF position.

e. Place a 115 volt AC test lamp with grounding type plug into each convenience and equipment receptacle of J3 thru J8, J11 thru J34.

f. Open (and secure open) outer exhaust port covers of blowers 1 and 2.

g. Place by-pass blackout switch S4 in the ON position.

h. Insert AC line plugs of exhaust blowers (1 and 2) into receptacles J9 and J10 respectively.

4.6.3.2 Line voltmeter and indicator lamp.-

a. Apply 115 volt AC (nominal), 60 cycles, power to power connector J1 in power entrance box. Power indicator neon lamp DS12 shall light; test lamp in receptacle of J3 shall light; and facility voltmeter in power distribution panel shall indicate power to the shelter.

b. Measure specific line voltage by connecting a calibrated external AC voltmeter between Pin 1 and Pin 2 of power connector J2.

c. Read voltage indicated on facility voltmeter. The voltage read shall be within 3% of the voltage read by 4.6.3.2b.

d. Place main circuit breaker CB8 in ON position.

e. Short circuit breaker CB13, test lamp in receptacle, J3 shall go out.

4.6.3.3 Circuit breakers and switches.-



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Circuit No.1.- (Lights circuit breaker CB1)

a. Place circuit breaker CB1 in the ON position. Neon lamp DS1 shall light.

b. Place fluorescent light switches S2 and S3 in ON position. All fluorescent lamps, DS13 thru DS20 shall light.

c. Place incandescent light switches S7 and S8 in ON position. Incandescent lamps DS21 thru DS24 shall light.

d. Place circuit breaker CB1 in OFF position. Neon lamp DS1 and all fluorescent and incandescent lamps, DS13 thru DS24 shall go out,

e. Place circuit breaker CB1 in ON position. With shelter door open, place by-pass blackout switch S4 in OFF position. All fluorescent and incandescent lamps, (DS13 thru DS24), shall go out.

f. Close shelter door. All fluorescent and incandescent lamps, DS13 thru DS24, shall light.

g. Open door and place by-pass blackout switch S4 in ON position.

Circuit No.2.- (Blowers circuit breaker CB2)

a. Place circuit breaker CB2 in ON position. Neon lamp DS2 shall light.

b. Place blower 1 switch S5 in ON position. Blower 1 shall operate.

c. Place blower 2 switch S6 in ON position. Blower 2 shall operate.

d. Place circuit breaker CB2 in OFF position. Neon lamp DS2 shall go out and blowers 1 and 2 shall stop.

Circuit No.3.- (Heater #1 circuit breaker CB3)

a. Place circuit breaker CB3 in ON position. Neon lamp DS3 shall light, and test lamp in receptacle J4 shall light.

b. Place circuit breaker CB3 in the OFF position. Neon lamp DS3 shall go out and test lamp in receptacle J4 shall go out.



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Circuit No.4.- (Heater #2 circuit Breaker CB4)

a. Repeat procedures 4.6.3.3 Circuit No.3 for Heater #2 circuit breaker CB4 and neon lamp DS4, and test lamp in receptacle J8.

Circuit No.5.- (Convenience circuit breaker CB5)

a. Place circuit breaker CB5 in ON position. Neon lamp DS5 shall light and test lamp in each receptacle of J5,J6 and J7 shall light.

b. Place circuit breaker CB5 in OFF position. Neon lamp DS4 shall go out. Test lamp in each receptacle of J5,J6 and J7 shall go out.

Circuit No.6.- (Regulator circuit breaker CB6)

a. Place circuit breaker CB6 in ON position. Neon lamp DS6 shall light. Place switch of overvoltage control unit in ON position neon lamp on this unit shall light.

b. Place circuit breaker CB6 in OFF position. Neon lamps DS6 and on overvoltage control shall go out.

Circuit No.7.- (Spare circuit breaker CB7)

a. Place circuit breaker CB7 in ON position. Neon lamp DS7 shall light.

b. Place circuit breaker CB7 in OFF position. Neon lamp DS7 shall go out.

Circuit No.8.- (Main circuit breaker CB8)

a. Place circuit breakers CB1 thru CB7 in ON position. All lamps DS1 thru DS7 and DS24 shall light. Blower 1, blower 2 shall operate. Test lamp in each receptacle of J4 thru J8 shall light.

b. Place circuit breaker CB8 in OFF position. All lamps DS1 thru DS7 and DS13 thru DS24 shall go out. Blower 1, blower 2 shall cease to operate. Test lamp in each receptacle of J4 thru J8 shall go out.



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c. Place circuit breakers CB1 thru CB5 in OFF position.

d. Place circuit breakers CB8 and CB6 in ON position.

Circuit No.9.- (System 1&2) circuit breaker CB9)

a. Place circuit breaker CB9 in ON position. Neon lamp DS8 shall light. Test lamp in each receptacle of J15 thru J19 shall light.

b. Place circuit breaker CB9 in OFF position. Neon lamp DS8 shall go out. Test lamp in each receptacle of J15 thru J19 shall go out.

Circuit No.10.- (System 3&4 circuit breaker CB10) Repeat procedures 4.6.3.3 Circuit No.9 for circuit breaker CB10, neon lamp, DS9 and receptacles J20 thru J24.

Circuit No.11.- ( System 5&6 circuit breaker CB11) Repeat procedure 4.6.3.3 Circuit No.9 for circuit breaker CB11 neon lamp, DS10, and receptacles J25 thru J29.

Circuit No.12.- (System 7&8 circuit breaker CB12) Repeat procedure 4.6.3.3 circuit No.9 for circuit breaker CB12, neon lamp DS11 and receptacles J30 thru J34.

Circuit No.13.- (Power entrance box circuit breaker CB13) Reset circuit breaker CB13 in ON position. Test lamp in receptacle J3 shall light.

OVERVOLTAGE CONTROL UNIT

a. Place circuit breakers CB6,CB9,CB10,CB11 and CB12 in ON position. Neon lamp DS6 shall light. Place switch in overvoltage control unit in on position. Neon lamp on overvoltage control unit shall light and neon lamps DS8 thru DS11 and test lamp in each receptacle J15 thru J34 shall light.

b. Place overvoltage control switch to OFF position. Neon lamp on overvoltage control unit, and neon lamps DS8 thru DS11 and test lamp in each receptacle J15 thru J34 shall go out. Place overvoltage control switch to ON position.

c. Place circuit breaker CB6 in OFF position. Neon lamp DS6 shall go out. Neon lamp on overvoltage control unit and neon lamps DS8 thru DS11 and test lamp on each receptacle J15 thru J34 shall go out.



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d. Place circuit breaker CB6 in ON position. Only neon lamp DS6 shall light.

e. Disconnect all test equipment and secure operational test.

4.6.4 Signal (audio) system wiring and components installation tests.-

4.6.4.1 Continuity and circuit resistance.- Continuity resistance which satisfy the requirements of 3.4.4.1 shall be made on each audio circuit indicated by the following paragraphs (using a low resistance continuity tester or equivalent producing a test current of not more than 0.010 amps.)

a. Prior to conducting signal wiring and components tests of 4.6.4.1, 4.6.4.2 and 4.6.4.3 all system cables shall be plugged into respective connectors on signal entrance boxes by systems and channels.

b. Place all cable/binding post channel switches, S1 thru S12 of system 1 thru 8, in BINDING POST position.

c. Connect tester between Pin A (of audio connector, equipment end system 1, channels 1-4) and Pin 2B (of audio connector J1 in signal entrance box) (see drawings SC-D-681007 and SC-D-681008) no circuit continuity shall exist.

d. Place channel switch S1 of system 1 to CABLE position. Circuit continuity shall exist and meet the requirements of 3.4.4.1. Replace channel switch S1 to BINDING POST position.

e. Disconnect tester leads from Pin A of audio connector and Pin 2B of audio connector J1.

f. Repeat procedures 4.6.4.1c thru e for each remaining conductor (Pin B thru Pin S of audio connector) to its corresponding pin of audio connector J1; each time placing to CABLE position the channel switch correlating with the circuit being tested and replacing to BINDING POST position following completion of test.

g. Repeat procedures 4.6.4.1c thru g for each conductor terminated at audio connectors of system 1 channels 5-12.

h. Repeat procedures 4.6.4.1b thru g for each conductor terminated in audio connector system 2 thru 8.



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i. Place all CABLE/BINDING POST channel switches, S1 thru S12 of system S1 thru 8, to CABLE position.

j. Connect tester between Pin A of audio connector and binding post 2B system 1, in signal entrance box. No circuit continuity shall exist.

k. Place channel switch S1 of system 1 to BINDING POST position. Circuit continuity shall exist and meet the requirements of 3.4.4.1. Replace channel switch S1 to CABLE position.

l. Disconnect tester leads from Pin A of audio connector, and BINDING POST 2B system 1.

m. Repeat procedure 4.6.4.1j thru l for each of the remaining conductor (Pin B thru Pin S of audio connector) to its corresponding binding post of system 1; each time placing to BINDING POST position the channel switch correlating with the circuit being tested and replacing to CABLE position following test.

n. Repeat procedures 4.6.4.1j thru m for each conductor terminated at audio connectors of system 1 channel 5-12.

o. Repeat procedures 4.6.4.1j thru n for each conductor terminated on audio connectors systems 2 thru 8.

p. Connect phone/intercom parallel interconnecting cable assembly between the phone/intercom plugs located on raceway and signal entrance boxes.

q. Circuit continuity that meets the requirements of 3.4.4.1 shall exist between the following points:

From tip of Phone Jack to:
Pin 25A of audio connector J1 thru J8.
Binding Post 25A of system 1 thru system 8.
Phone A binding post at power entrance box.

From sleeve of Phone Jack to:
Pin 25B of audio connector J1 thru J8.
Binding Post 25B of system 1 thru system 8.
Phone B binding Post at power entrance box.

From tip of Intercom Jack to:
Pin 26A of audio connector J1 thru J8.
Binding Post 26A of system 1 thru system 8.
Intercom A binding post at power entrance box.



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From sleeve of Intercom Jack to:
Pin 26B of audio connector J1 thru J8.
Binding Post 26B of system 1 thru system 8.
Intercom B Binding Post at power entrance box.

r. Disconnect phone/intercom cable installed in 4.6.4.1p above from raceway only.

4.6.4.2 Dielectric strength of insulation.- Dielectric strength of insulation tests which satisfy the requirements of 3.4.4.2 shall be made for each audio circuit indicated by the following paragraphs (Method 301 of MIL-STD-202 shall be used utilizing a non-destructive dielectric test instrument and suitable test jigs or an automatic tester may be used subject to government approval):

a. Place all CABLE/BINDING POST channel switches, S1 thru S12 of system 1 thru 8 in the CABLE position.

b. Connect the following to facility ground terminal:

(1) Pins 1A thru 26B of audio connector J1 thru J8.

(2) Binding posts 1A thru 26B of system 1 thru system 8.

(3) Shields of audio connectors of system 1 thru 8.

c. Connect tester between the facility ground terminal and Pin A of audio connector channels 1 thru 4. Remove ground connection from Pin 2B of audio connector J1. (See drawings SC-D-681007 and SC-D-681008)

d. Adjust tester to 600 volts DC output and maintain this potential for a continuous period of not less than 5 seconds the requirements of 3.4.4.2 shall be met.

e. Disconnect tester lead from Pin A of audio connector channel 1 thru 4 and re-establish ground connection to Pin 2B of audio connector J1.

f. Repeat procedures 4.6.4.2c thru e for each remaining conductor (Pin B thru Pin S of audio connector channels 1 thru 4 each time ungrounding the pin of audio connector J1 correlating with the circuit being tested and reconnecting pin to facility ground terminal following completion of the test.

g. Repeat procedures 4.6.4.2c thru f for each conductor terminated on audio connectors of system 1 channel 5-12.

h. Repeat procedures 4.6.4.2c thru g for systems 1 thru 8.



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i. Place all CABLE/BINDING POST channel switches, S1 thru S12 of system 1 thru 8 in the BINDING POST position.

j. Repeat procedures 4.6.4.2c thru h substituting BINDING POST of system 1 thru 8 for audio connector J1 thru J8 respectively.

k. In power entrance box connect phone and intercom BINDING POST to facility ground terminal.

l. Remove ground from BINDING POST A of Phone circuit and connect tester between facility ground terminal and BINDING POST A.

m. Adjust tester for 200 VDC output and maintain this potential for a period of not less than 5 seconds the requirements of 3.4.4.2 shall be met.

n. Repeat procedures 4.6.4.2 l and m for each remaining conductor, each time ungrounding BINDING POST correlating with circuit being tested and reconnecting BINDING POST to facility ground terminal following completion of test.

NOTE: To eliminate danger of shock: Discharge conductors following each test by connecting a 500 ohm wirewound resistor (or equivalent test lamp) to the pins tested.

4.6.4.3 Insulation resistance.- Insulation resistance which satisfy the requirements of 3.4.4.3 shall be made on each audio circuit indicated by the following paragraphs (Method 302 of MIL-STD-202 shall be used, utilizing a megohmmeter or equivalent tester, and suitable test jig or an automatic tester subject to government approval):

a. Repeat procedures 4.6.4.2a thru j, except 4.6.4.2d which shall be replaced by 4.6.4.3b below.

b. Adjust tester to 500 volts DC output. The requirements of 3.4.4.3 shall be met.

NOTE: To eliminate danger of shock: Discharge conductors following each test by connecting a 500 ohm wirewound resistor (or equivalent test lamp) to the pins tested.

4.6.5 Video system wiring and components installation.-

4.6.5.1 Continuity and circuit resistance.- Continuity resistances which satisfy the requirements of 3.4.5.1 shall be made on each video circuit indicated by the following paragraph (using a low resistance continuity tester):



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a. Connect tester between inner conductor of IN coaxial cable connector of system 1 in video entrance box and inner conductor of female connector of cable assembly marked VIDEO SYS-1. Circuit continuity shall exist and meet the requirements of 3.4.5.1.

b. Connect tester between outer conductors (Shields) of circuit under test. Circuit continuity shall exist and meet the requirements of 3.4.5.1.

c. Connect tester between shield of circuit under test and facility ground terminal. Circuit continuity shall exist and meet the requirements of 3.4.5.1.

d. Connect tester between inner conductor of OUT coaxial cable connector of system 1 in video entrance box and inner conductor of male connector of cable assembly marked VIDEO SYS-1. Circuit continuity shall exist and meet the requirements of 3.4.5.1.

e. Repeat procedures 4.6.5.1b and c.

f. Repeat procedures 4.6.5.1a thru e for IN and OUT coaxial cable connectors of system 2 thru 8.

4.6.5.2 Dielectric strength of insulation.- Dielectric strength of insulation tests which satisfy the requirements of 3.4.5.2 shall be made on each video circuit indicated by the following paragraphs (Method 301 of MIL-STD-202 shall be used utilizing a non-destructive dielectric tester.):

a. Position or separate the unconnected video connectors so that none are shorted to each other or to facility ground.

b. Remove all lightning protector elements.

c. Connect tester between inner conductor of IN coaxial cable connector of system 1 in video entrance box and the facility ground terminal.

d. Adjust tester to 1700 volts DC output and maintain this potential for a continuous period of not less than 30 seconds. The requirements of 3.4.5.2 shall be met.

e. Disconnect tester lead from inner conductor of IN coaxial cable connector of system 1.

f. Repeat procedures 4.6.5.1c thru e for inner conductor of OUT coaxial cable connector of system 1; and for inner conductor of IN and OUT coaxial cable connectors of system 2 thru 8.



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g. Replace all lightning protector elements removed by 4.6.5.2b.

h. Connect tester between inner conductor of IN coaxial cable connector of system 1 in video entrance box and the facility ground terminal.

i. Adjust tester to 1200 volts DC output and maintain this potential for a continuous period of not less than 30 seconds. The requirements of 3.4.5.2 shall be met.

j. Disconnect tester lead from inner conductor of IN coaxial cable connector of system 1.

k. Repeat procedures 4.6.5.2h thru j for inner conductor of OUT coaxial cable connector of system 1; and for inner conductor of IN and OUT coaxial cable connectors of system 2 thru 8.

NOTE: To eliminate danger of shock: Take preventive measures that will preclude personnel coming in contact with connectors inside the shelter during performance of tests.

4.6.5.3 Insulation resistance.- Insulation resistance tests which satisfy the requirements of 3.4.5.3 shall be made on each video circuit indicated by the following paragraphs (Method 302 of MIL-STD-202 shall be used utilizing a megohmmeter or equivalent tester):

a. Repeat procedures 4.6.5.2a thru f except 4.6.5.2d which shall be replaced by 4.6.5.3b below.

b. Adjust tester to 1000 volts DC output and maintain this potential for a continuous period of not less than 30 seconds. The requirements of 3.4.5.3 shall be met.

NOTE: To eliminate danger of shock: Take preventive measures that will preclude personnel coming into contact with connectors inside the shelter during performance of tests.

4.7 System test.- Install operational equipment in their respective rack locations and connect power and signal cables to designated receptacles and connectors. Interconnect TD-204()/U multiplexers (back to back operation). For cabling of equipment See Figure 3.

a. Place all CABLE/BINDING POST channel switches, S1 thru S12 of systems 1 thru 8 in BINDING POST position at Signal Entrance Boxes.



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b. Perform the following checks on the Converter Telephone Signal CV-154β()/G:

(1) Check to see that the POWER switch is operated to OFF.

(2) Operate the CH-2W-4W switch to 2W and the signaling mode switches to AC.

(3) Operate the POWER switch to ON.

(4) Check the 20 cycle indicator on panel 18A2 is glowing.

(5) Operate the meter selector switch through -,+,20 cycle drive, and 20 cycles, and check for a yellow area indication on the TEST ALIGN meter.

(6) Operate the meter selector switch to 1600 cycles and check for a green area indication on the TEST ALIGN meter. If necessary, adjust the ADJ 1600 control on panel 18A2 for a center indication in the green area.

c. Perform the following procedures on each TD-352()/G:

(1) Check to see that the AC POWER switch is operated to OFF.

(2) Press the PUSH TO RELEASE CHASSIS button and pull the TD-352 far enough out of its case to expose the service facility panel.

(3) Operate the CHAN switch to OFF.

(4) Operate the AUX switch to OUT.

(5) Operate the ADDRESS switch to MASTER position.

(6) Operate 2W-4W switch to 2W position.

(7) Operate the METER SELECT switch to SERV FAC.

(8) Operate the AC POWER switch to ON.

(9) Operate the SERV SEL switch through +25,+10,+45,-4.5 and -12, and check for a yellow area indication on the TEST ALIGN meter.

(10) Operate the SERV SEL switch to -5.2 and check for a green area indication on the TEST ALIGN meter. Operate the SERV SEL switch to +25 and return TD-352 to its case.



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d. Perform the following procedures on each TD-204/U connected in a system.

(1) Operate the AC POWER, and TALK-OFF-SIG switches to OFF.

(2) Depress the PUSH TO RELEASE CHASSIS button and slide the TD-204/U out of its case far enough to expose the service facility panel and panel 6A4.

(3) Operate the TRAFFIC SEL switch to 12 Channel position.

(4) Operate the NORM OPR-ZERO SET-READ switch to NORM OPR.

(5) Operate both MILES switches to 0.

(6) Operate the TONE-OFF switch on panel 6A2 to OFF.

(7) Operate the MILE switch on panel 6A4 to ½ and on panel 6A5 to ½.

(8) Operate the AC POWER switch to ON. Check to see that the AC POWER and ALARMS NO CABLE CURRENT indicators light and the buzzer sounds. Silence the ALARMS BUZZER OFF switch.

(9) Operate the CABLE POWER switch to OFF.

(10) Operate the METER SELECT switch to SERV FAC.

(11) Operate serv.sel switch -10,+10, sum ± 3 and bal, and check for a yellow area indication on the test align meter for each position. Operate SERV SEL switch to +10 return TD-204 to case.

e. Connect a Telephone Set TA-312()PT phone 1 to BINDING POST pair 2WT/R Channel 1 System 1, signal entrance box roadside.

f. Connect a telephone set TA-312()PT phone 2 to BINDING POST pair 2WT/R channel 1 system 5, signal entrance box curbside.

g. Ring phone 2 from phone 1 ring phone 1 from phone 2 establish communications between phones.

h. Repeat step 4.7 g above with phones connected to 2WT/R for the remaining channels of systems 1&5.

i. Set 2 wire-4 wire switches in Converter Telephone Signal CV-1548()/U and Multiplexer TD-204()/U in 4 wire position.



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j. Connect Phone 1 to BINDING POST pair 2WT/R channel 1 System 1, Connect Phone 2 to BINDING POST pair 4WT, channel 1, system 5. Receive communications at Phone 1 from Phone 2.

k. Connect Phone 1 to BINDING POST pair 4WT, channel 1, system 1, connect phone 2 to BINDING POST pair 2WT/R channel 1, system 5, receive communications at Phone 2 from Phone 1.

l. Repeat 4.7 (j)(k) above for remaining channels of system 1 and 5.

m. With the following systems cabled per FIGURE 3, repeat 4.7 (e) thru (j) for:

System 2 and System 6
System 3 and System 7
System 4 and System 8

n. Turn off all equipment and disconnect all cable connections and secure operational test.

4.8 Vehicular transportation test.- The facility shall be secured (and blocked to prevent shifting) to a government furnished 2½ ton M-35, truck and transported over the specially prepared course described in 4.8.1 for 10 laps at a speed of 5 mph, 10 laps at a speed of 10 mph, 10 laps at a speed of 15 mph, and 10 laps at a speed of 18-20 mph. A lap is defined as traversing the course in one direction. The facility shall meet the requirements of paragraph 3.6 after completion of the test. (Note: The facility shall be loaded to simulate the AN/TCC-61 using live and dummy loads (See 6.2(b) for this test).

4.8.1 Vehicular transportation course.- The course for the military truck transportation shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1-inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be anchored securely by any convenient means. The ground on which the timbers are placed shall be hard and level.



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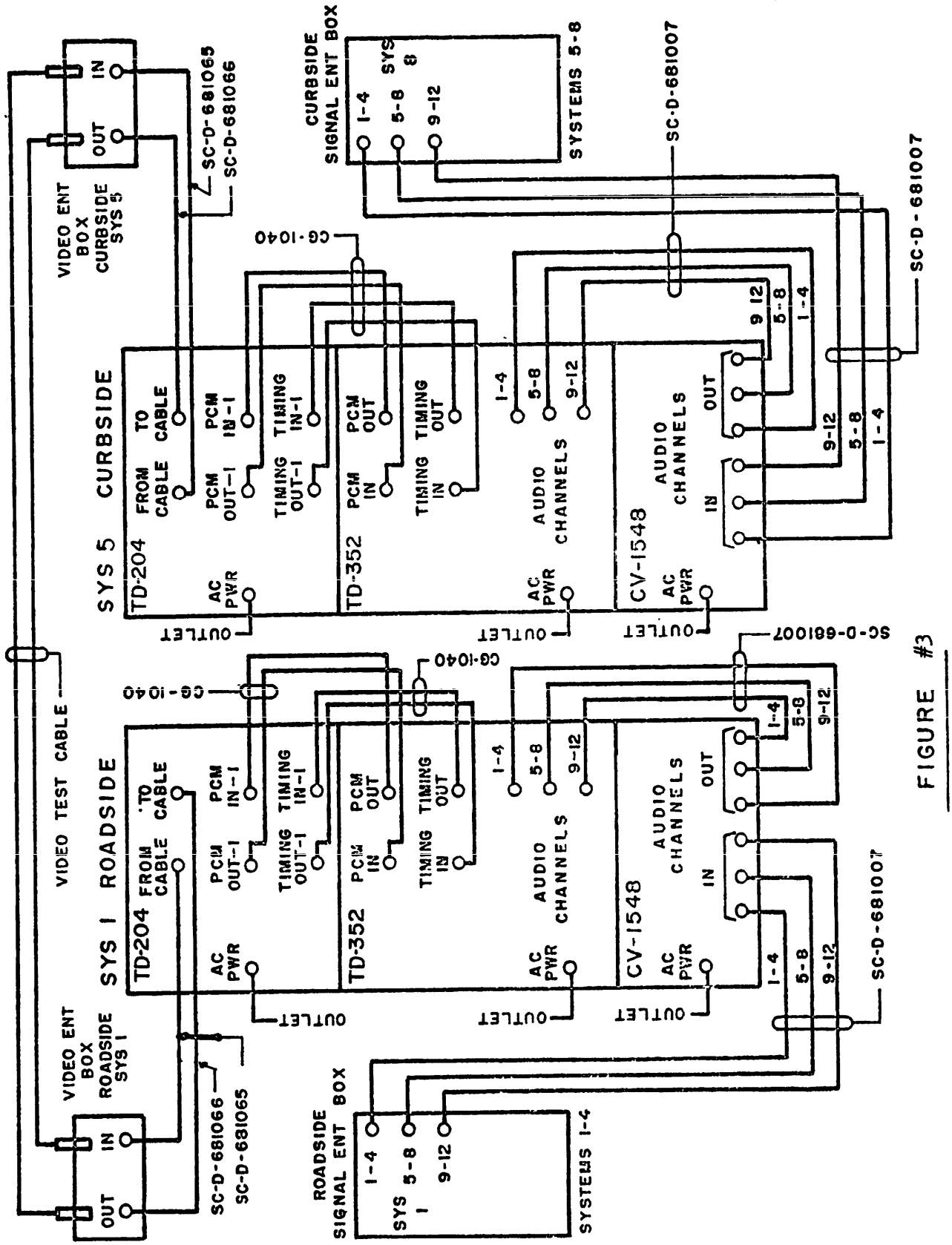



FIGURE #3



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4.9 Rail transport, impact test.- The shelter facility, with loads as specified in 4.7, shall be loaded onto a railroad flatcar in accordance with Figure 4. The test shall be conducted on a flat stretch of track. A loaded flatcar totaling 165,000 pounds (load on flatcar shall be constrained to prevent shifting during this test) traveling at 9 miles per hour (or equivalent conditions approved by the Government) shall be impacted against the stationary test flat car coupled to two empty boxcars, all with brakes off. Four impacts shall be performed, two impacts into each end of the test flatcar. The cabling or blocking holding the shelters in place shall be tight at the start of each impact run. If the cabling or blocking is torn loose by the impact, the run shall be repeated. A damage survey shall be made after each impact and the facility shall meet the requirements of 3.7.

4.10 Drop test.-

4.10.1 Flat drop.- The shelter facility loaded as in 4.8 shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.8.

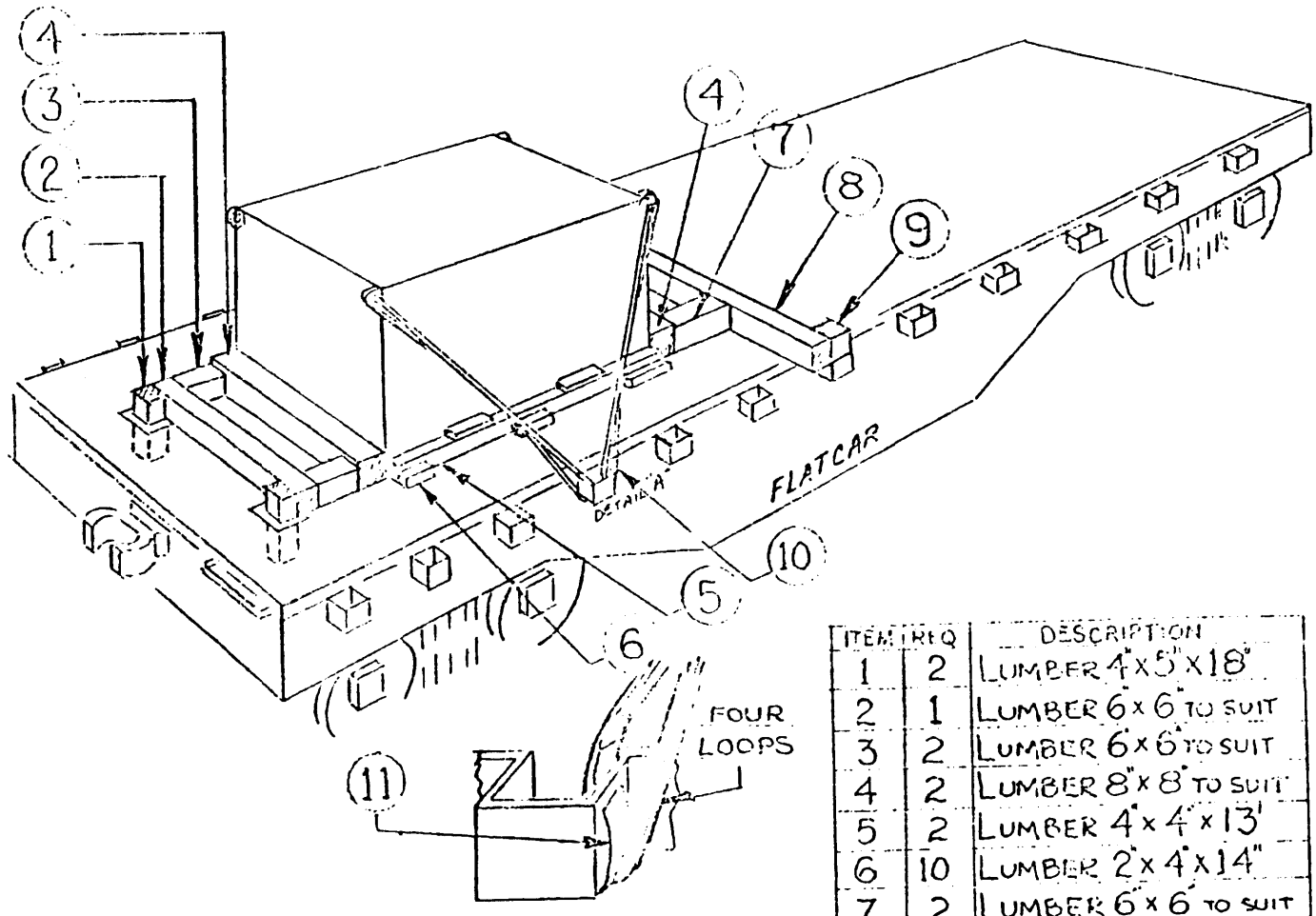
4.10.2 Rotational drop.- The shelter facility loaded as in 4.7, shall have one skid (or the knuckles of the skids) resting on a nominal 4 inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.8.

4.11 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc. shall be subjected to the following two part test (Note: Perform tests with inside cover of entrance boxes removed):

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G 29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using nine nozzles, and the roof using four nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches.



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DETAIL "A"

Wire tiedown same for all points of contact with stake pockets

| ITEM | REQ | DESCRIPTION |
|---|---|---|
| 1 | 2 | LUMBER 4"x5"x18" |
| 2 | 1 | LUMBER 6"x6" TO SUIT |
| 3 | 2 | LUMBER 6"x6" TO SUIT |
| 4 | 2 | LUMBER 8"x8" TO SUIT |
| 5 | 2 | LUMBER 4"x4"x13' |
| 6 | 10 | LUMBER 2"x4"x14" |
| 7 | 2 | LUMBER 6"x6" TO SUIT |
| 8 | 1 | LUMBER 6"x6" TO SUIT |
| 9 | 2 | LUMBER TO SUIT |
| 10 | 16 | WIRE #8 SOLID, HARD, 2050 LB TEST |
| 11 | 4 | FLEX. METAL SHEETING |

NOTES:

a. The blocking lumber shall be notched to accomodate towing eyes, bolts and rivet heads.

b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.





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(See FIGURE 5 for nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4,12, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door-end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.9. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

b. Repeat the test of 4.11a with covers open to their normal operating position.

4.12 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 21 inches (measured from the bottom of the skids) and meet the requirements of 3.10. No special fording kits shall be used. The shelter shall be restrained by external tiedowns. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.13 Mounting insert tests.-

4.13.1 Preproduction insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility.

4.13.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility.

Table V - Mounting Inserts Loads

| Insert size | Preproduction test | | Production Test | |
|---|---|---|---|---|
| | Axial(lbs.) | Torque (in.-lbs.) | Axial(lbs.) | Torque(in.-lbs.) |
| 5/16-18 | 2000 | 100 | 1000 | 100 |
| 1/4-20 | 1300 | 60 | 650 | 60 |

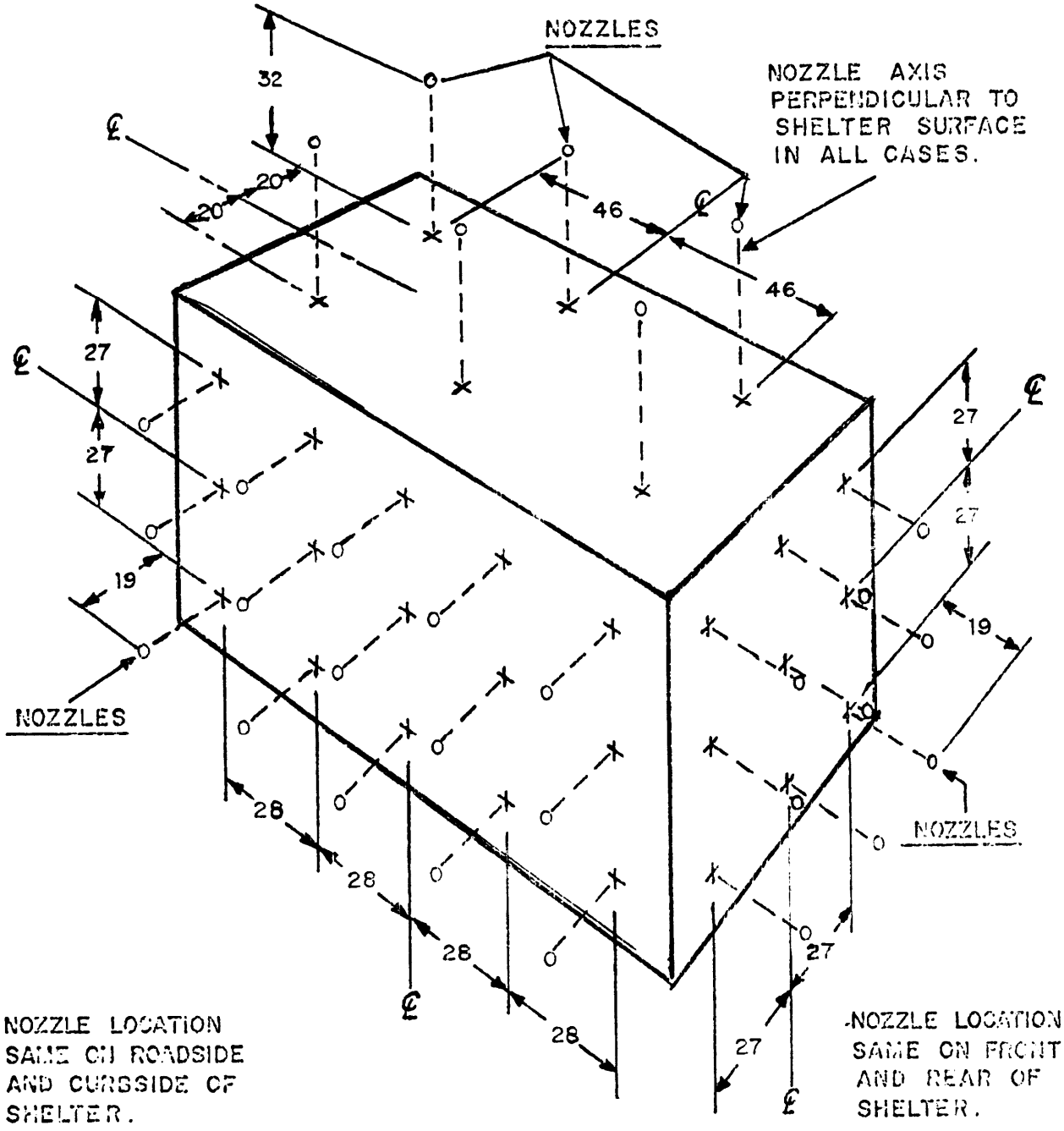
NOTE: Torque loads applied through bolt whose head holds a stationary plate against head of insert.



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FIGURE 5

NOZZLE LOCATION FOR TEST PER 4.11




*Note :*

*GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.*



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4.14 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.21.4.

4.15 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.22. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks and cabinets.

4.16 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.23 and Table VI.

Table VI - Classification of Visual and Mechanical Defects

| Classification | Defects |
|---|---|
| Major | 1. Mounting plates missing or mislocated.<br>2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible.<br>3. Welds - cracked or porous.<br>4. Sealer improperly applied (holes, separations, or lack of adhesion).<br>5. Mounting inserts missing or mislocated.<br>6. Burrs or sharp edges in wire duct not removed.<br>7. Equipments or bracketry loose or missing.<br>8. Doors, etc. inoperative.<br>9. Power receptacles, switches, jacks, wire ducts or other electrical components loose, improperly located or inoperative.<br>10. Locking or holding devices missing or inoperative.<br>11. Gaps between sections of wire duct exceed 1/16". |
| Minor | 1. Doors, not easily operable.<br>2. Fastening devices difficult to operate.<br>3. Finish - abrasions or scratches. |



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4.17 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.18 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

| | |
|---|---|
| Paragraph 4.6.3,4.6.4.1,4.6.5.1 | Power and Signal wiring |
| Paragraph 4.16 | Visual and mechanical |

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the shelter facility shall be in accordance with applicable requirements of Specification MIL-P-55505.

6. NOTES

6.1 Intended use.- The Shelter, Electrical Equipment S-301()/TCC-61 is the shelter facility portion of the AN/TCC-61() and it is used to house the operational equipments including:

a. 8 ea TD-352()/U Multiplexer

b. 8 ea TD-204()/U Multiplexer

c. 8 ea CV-1548()/G Converter, Telephone Signal

d. 1 ea CN-514()/GRC Voltage Regulator

e. 1 ea TA-312()/PT Telephone

f. 2 ea H-91()/U Headset-Microphone

g. 1 ea LS-147()/FI Intercom

h. 8 ea Security Equipment (Optional)

forming the telephone terminal.



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6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Preproduction and Group C inspection of the shelter facility required use of contractor fabricated dummy loads for the communications security equipments (8 ea) and government furnished live loads (converters and multiplex components) unless otherwise specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-301()/TCC-61.

d. A list of equipments to be government loaned for testing purposes.

e. Submission of the statement referenced in 3.19.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-301R/TCC-61 The contractor should apply for nomenclature in accordance with the applicable clause in the contract.



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6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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Preparing Activity
Army - El

Project No. 5805-A129



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-S-55646(EL) SHELTER, ELECTRICAL EQUIPMENT S-301/TCC-61

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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